

Multi-Nutrient Personalized Food Recommendations

Kevin Sanchez, Kevin Munoz, and Jose Acevedo

Abstract

Diet plays a major role in managing hypertension, yet most personalized food recommendation systems treat a user's health condition as one more soft, weighted preference alongside taste and convenience rather than as a firm clinical requirement. As a result, a food that violates an established dietary guideline can still be ranked highly if it scores well on other nutrients. This project extends an existing unsupervised food-clustering approach, which groups foods by nutrient similarity to generate personalized recommendations without requiring any user rating history, with a hard constraint layer for users managing hypertension. Sodium limits drawn from the Dietary Approaches to Stop Hypertension (DASH) eating plan are enforced before nutrient-similarity ranking takes place, so that foods exceeding the guideline are excluded or substituted rather than merely down-weighted. Unlike prior systems that enforce comparably strict clinical rules through a hand-built clinical ontology validated by practicing physicians, this approach relies on a single, publicly documented numeric guideline, keeping the system reproducible and feasible to build and evaluate within a single academic semester. The system will be evaluated on a set of synthetic hypertensive user profiles, comparing recommendation quality and guideline-violation rates against a soft-weighted baseline. This work is intended to demonstrate that a lightweight, guideline-derived hard constraint can meaningfully improve the clinical reliability of an existing, easily reproducible recommendation pipeline, addressing a gap identified across multiple independent reviews of food recommendation research.

Index Terms

food recommendation systems, personalized nutrition, recommender systems, literature review, dietary recommendation.

I. LITERATURE REVIEW

A. Systematic and Scoping Reviews

Two broad reviews establish the current landscape of health-oriented food recommendation. Dong et al. [1] conducted a PRISMA-ScR scoping review of fifteen diet-related health recommender systems for patients with chronic conditions, published between 2010 and 2024. They found that diabetes (60%) and hypertension (40%) were the most commonly targeted conditions, that 87% of the reviewed systems used a hybrid recommendation approach combining hard constraints with preference-based weighting, and that only one of the fifteen systems adapted its recommendations over time. Critically, none of the reviewed systems validated whether their recommendations were medically sound, and few considered food-medication interactions. Yera et al. [2] performed a complementary PRISMA-2020 systematic review focused specifically on diabetic patients, analyzing 34 papers grouped into four technical categories: semantic/ontology-based, optimization-based, rule-based/classification, and interaction-based approaches. Across all four categories, the review identified recurring weaknesses, including a scarcity of diabetes-specific datasets, inconsistent or absent evaluation protocols, and a lack of a shared research framework that would allow systems to be meaningfully compared. Together, these two reviews confirm that health-condition awareness in existing food recommenders is generally soft, inconsistently evaluated, and rarely validated for clinical safety.

B. Clustering and Multi-Objective Optimization

Sarker and Tanjim [3] proposed a multinutrient clustering framework that groups 8,790 USDA foods into eight clusters using k -means over 23 nutritional attributes, allowing users to personalize recommendations through nutrient sliders and preset dietary profiles. Evaluated on simulated queries, their approach outperformed popularity-based and single-nutrient-filtering baselines by 46.4% and 273%, respectively, while requiring no user rating history. However, the system recommends single foods only, uses USDA data exclusively, and includes no explicit health-condition modeling. Huang et al. [4] extended this idea by combining collaborative filtering with an improved Multi-Objective Evolutionary Algorithm Based on Decomposition (MOEA/D) to jointly balance nine nutrients across a combination of eight recommended foods for a single day. Their improved algorithm (FNR-MOEA/D) achieved a lower average nutrient error than five competing multi-objective algorithms, demonstrating that multi-objective optimization can balance several nutrients across a food combination simultaneously. This work is limited to a single day, a fixed number of foods, and, like [3], includes no clinical constraints or adaptive feedback.

C. Knowledge Graphs and Graph Neural Networks

Chen et al. [5] introduced FKGM, a health-aware food recommendation model built on a collaborative recipe knowledge graph and trained with multi-task learning to jointly optimize for user preference and a healthiness score based on four nutrients (sodium, fat, sugar, saturated fat). FKGM outperformed four knowledge-graph baselines specifically on the health-alignment

task, though the authors note that only four nutrients were considered and that no user study was conducted. Ma et al. [6] proposed a related but distinct nutrition-related knowledge graph neural network (NRKG) that links recipes by nutrient-profile similarity to increase recommendation diversity and reduce repetitive suggestions, outperforming six graph neural network baselines across five ranking metrics. Both works explicitly identify the absence of allergy and dietary-restriction handling as a limitation and area for future work, reinforcing that even recent graph-based methods treat health as a soft, narrowly-scoped signal rather than a hard constraint.

D. Large Language Model-Based Approaches

Three works apply large language models to food recommendation in substantially different ways. Khamesian et al. [7] proposed NutriGen, which prompts ten different LLMs (including GPT and Llama variants) to directly generate a full day’s meal plan against a personalized nutrition database cross-checked with USDA data; the best-performing models (Llama 3.1 8B and GPT-3.5 Turbo) achieved calorie-target errors of 1.55% and 3.68%, respectively, though all user profiles were synthetic and no health-condition awareness was included. Papastratis et al. [8] instead used a variational autoencoder and a recurrent decoder, trained with loss functions that explicitly penalize deviation from EFSA and WHO nutritional guidelines, to generate full weekly meal plans conditioned on a user profile that includes cardiovascular disease, type-2 diabetes, and iron-deficiency flags; ChatGPT was used only afterward, to expand the meal database with nutritionally equivalent meals in other cuisines. In a direct comparison, this guideline-constrained approach outperformed ChatGPT prompted directly as a meal-plan recommender by roughly 12% in macronutrient accuracy and reduced average caloric deviation from approximately 17% to 0%, suggesting that grounding an LLM in explicit nutritional guidelines yields more reliable recommendations than prompting alone. Deng and Tu [9] proposed NutriRec, a multimodal LLM framework that combines textual, ingredient, image, and nutrient information with a sequential collaborative-filtering model of user behavior, enabling explainable “nutritional compromise” recommendations (e.g., substituting ingredients to reduce sodium while preserving a preferred flavor profile) validated across three separate datasets. Across all three works, health constraints remain soft or coarse-grained rather than clinically validated rules.

E. Large-Scale Collaborative Filtering on Real Cohort Data

Silva et al. [10] evaluated user-based and item-based collaborative filtering for dietary recommendation using real, non-simulated data from 12,667 participants of the ELSA-Brasil cohort study, achieving 88–91% precision in predicting food-group adherence. This work demonstrates that collaborative filtering can be validated on a large real-world population and a food catalog beyond USDA, but the authors note that the Food Frequency Questionnaire used cannot capture allergies or personal restrictions, and that results are specific to a Brazilian, non-clinical population.

F. Clinical Decision Support

Spoladore et al. [11] took a markedly different approach, developing OnT2D-DSS, an ontology-based clinical decision support system that encodes expert nutritional-therapy rules for type-2 diabetes to help family doctors generate explainable, patient-specific macro- and micronutrient targets. The system was validated in two preliminary studies with real nutrition experts and family doctors, receiving favorable usability ratings. OnT2D-DSS represents the only work surveyed here that enforces genuinely hard, clinically-derived constraints; however, it is doctor-facing rather than end-user-facing, is purely rule-based with no machine-learning or optimization component, and required substantial, ongoing clinical-expert involvement to build and validate.

G. Summary of Gaps

Considered together, these eleven works make clear that no existing system combines (i) recommendation at the level of full meals or full-day plans, (ii) real, clinically-derived health constraints rather than soft preference weighting, (iii) food data broader than a single national or single-source database, and (iv) adaptation based on user feedback over time. Full-day planning is demonstrated by [7], [4], and [8], each using a different underlying technique (LLM prompting, multi-objective optimization, and a guideline-constrained generative network, respectively). Hard clinical constraints are demonstrated only by [11], which is not paired with any of the recommendation techniques above. Broader dataset coverage is demonstrated by [10], [9], and, to a lesser extent, [8]. Adaptive, feedback-driven recommendation is demonstrated only partially, by [9] and by a single system within the fifteen reviewed in [1]. This gap analysis will directly inform the specific contribution of our project, described in Section II.

II. PROPOSED NOVEL CONTRIBUTION

Building on the gap analysis in Section I, we propose to extend the multinutrient clustering framework of Sarker and Tanjim [3] with a hard, guideline-derived constraint layer for a single chronic condition, hypertension, rather than continuing to treat health as one more soft-weighted nutrient preference as in [5] and [6]. Specifically, sodium limits drawn from the Dietary

Approaches to Stop Hypertension (DASH) eating plan [12] (2,300 mg/day, or 1,500 mg/day under the lower-sodium tier) will be enforced as a hard filter applied before the existing nutrient-similarity ranking, so that a food violating the sodium threshold is excluded or substituted rather than merely down-weighted alongside the other nutrients already used for personalization.

This design positions the project between the two extremes identified in Section I: the machine-learning-driven but softly-constrained systems of [3], [5], and [6], and the clinically rigorous but doctor-facing, non-adaptive, ontology-based OnT2D-DSS of [11], which required sustained collaboration with practicing clinicians and nutrition experts to build and validate. Because the DASH sodium threshold is a single, well-documented, publicly available numeric guideline, rather than the multi-condition clinical ontology underlying [11], this constraint can be implemented and evaluated without clinical-expert collaboration or patient recruitment, keeping the project within the scope of a one-semester capstone while still enforcing a genuinely hard, guideline-derived rule rather than a soft preference.

The system will be evaluated on a set of synthetic hypertensive user profiles, following the simulated-query evaluation methodology already used by [3] and [8]. The primary comparison will be between the proposed hard-constrained recommender and the original soft-weighted baseline, measured by (i) the rate at which each system recommends foods that violate the DASH sodium threshold, and (ii) any resulting change in overall recommendation quality relative to the target nutrient profile. As time permits, a second condition, such as an ADA-style carbohydrate guideline for type-2 diabetes, may be added to demonstrate that the approach generalizes beyond a single condition, consistent with the two conditions, hypertension and diabetes, that dominate the systems reviewed in [1].

REFERENCES

- [1] X. Dong, B. Yun, A. Pakarinen, Z. Zheng, H. Niu, T. Jin, C. Yuan, and J. Wang, "Diet-related health recommender systems for patients with chronic health conditions: Scoping review," *J. Med. Internet Res.*, vol. 28, p. e77726, 2026, doi: 10.2196/77726.
- [2] R. Yera, A. A. Alzahrani, L. Martínez, and R. M. Rodríguez, "A systematic review on food recommender systems for diabetic patients," *Int. J. Environ. Res. Public Health*, vol. 20, no. 5, p. 4248, 2023, doi: 10.3390/ijerph20054248.
- [3] R. Sarker and K. F. H. Tanjim, "A multinutrient clustering framework for personalized food recommendations," *Int. J. Food Sci.*, 2026, doi: 10.1155/ijfo/9969543.
- [4] S. Huang, C. Wang, and W. Bian, "A hybrid food recommendation system based on MOEA/D focusing on the problem of food nutritional balance and symmetry," *Symmetry*, vol. 16, no. 12, p. 1698, 2024, doi: 10.3390/sym16121698.
- [5] Y. Chen, Y. Guo, Q. Fan, Q. Zhang, and Y. Dong, "Health-aware food recommendation based on knowledge graph and multi-task learning," *Foods*, vol. 12, no. 10, p. 2079, 2023, doi: 10.3390/foods12102079.
- [6] W. Ma, M. Li, J. Dai, J. Ding, Z. Chu, and H. Chen, "Nutrition-related knowledge graph neural network for food recommendation," *Foods*, vol. 13, no. 13, p. 2144, 2024, doi: 10.3390/foods13132144.
- [7] S. Khamesian, A. Arefeen, S. M. Carpenter, and H. Ghasemzadeh, "NutriGen: Personalized meal plan generator leveraging large language models to enhance dietary and nutritional adherence," in *Proc. 47th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2025, doi: 10.1109/EMBC58623.2025.11253879.
- [8] I. Papastratis, D. Konstantinidis, P. Daras, and K. Dimitropoulos, "AI nutrition recommendation using a deep generative model and ChatGPT," *Sci. Rep.*, vol. 14, p. 14620, 2024, doi: 10.1038/s41598-024-65438-x.
- [9] X. Deng and W. Tu, "Personalized nutrition-aware dietary recommendation with multimodal large language models," *IEEE Access*, vol. 14, pp. 21791–21804, 2026, doi: 10.1109/ACCESS.2026.3660702.
- [10] V. C. Silva, B. Gorgulho, D. M. Marchioni, S. M. Alvim, L. Giatti, T. A. de Araujo, A. C. Alonso, I. de Souza Santos, P. A. Lotufo, and I. M. Bensenor, "Recommender system based on collaborative filtering for personalized dietary advice: A cross-sectional analysis of the ELSA-Brasil study," *Int. J. Environ. Res. Public Health*, vol. 19, no. 22, p. 14934, 2022, doi: 10.3390/ijerph192214934.
- [11] D. Spoladore, F. Stella, M. Tosi, E. C. Lorenzini, and C. Bettini, "A knowledge-based decision support system to support family doctors in personalizing type-2 diabetes mellitus medical nutrition therapy," *Comput. Biol. Med.*, vol. 180, p. 109001, 2024, doi: 10.1016/j.combiomed.2024.109001.
- [12] National Heart, Lung, and Blood Institute, "DASH eating plan," U.S. Dept. of Health and Human Services, 2021. [Online]. Available: <https://www.nhlbi.nih.gov/education/dash-eating-plan>